

Please write clearly in block capitals.

Centre number

--	--	--	--	--

Candidate number

--	--	--	--

Surname

--

Forename(s)

--

Candidate signature

--

INTERNATIONAL GCSE BIOLOGY

Paper 2

Monday 12 November 2018 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a ruler with millimetre measurements
- a scientific calculator.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	



Answer **all** questions in the spaces provided.

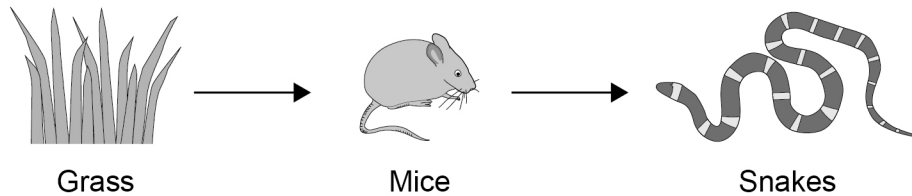
Do not write
outside the
box

0 1

A food chain shows the feeding relationship between organisms.

Figure 1 is a food chain.

Figure 1



0 1 . 1

Grass carries out photosynthesis.
What is the source of energy for photosynthesis?

[1 mark]

0 1 . 2

Suggest **two** possible effects on this food chain if a lot of the mice died.

[2 marks]

1 _____

2 _____

Biomass is the mass of material in living organisms.

Table 1 shows the total biomass at each stage in this food chain in one field.

Table 1

Organism	Total biomass in kg
Grass	900
Mice	90
Snakes	20



0 1 . 3

Calculate the total biomass lost from this food chain.

[1 mark]

Total biomass lost = _____ kg

0 1 . 4

Sketch a pyramid of biomass for the food chain in **Figure 1**.

Label the pyramid.

[2 marks]

0 1 . 5

Give **two** ways biomass can be lost from this food chain.

[2 marks]

Tick (✓) **two** boxes.Grass photosynthesising ☐Grass reproducing ☐Mice growing ☐Mice producing faeces
(droppings) ☐Mice respiring ☐

Turn over ►



0 2

Tapeworms live in the intestine of another animal. This animal is called the host.

0 2 . 1

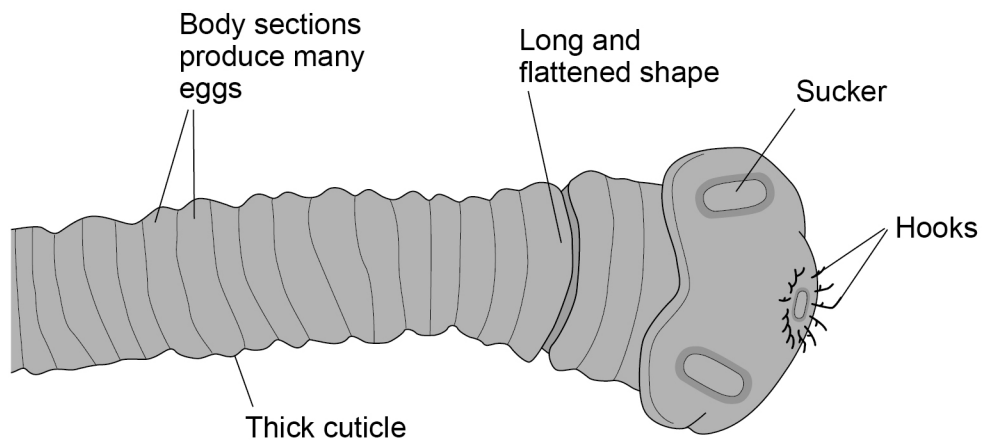
What name is given to the tapeworm in this relationship?

[1 mark]

The tapeworm absorbs food through its skin from the intestine of the host animal.

Figure 2 shows part of a tapeworm and the features which help it to survive in the intestine of its host.

Figure 2



0 2 . 2

Draw **one** line from each feature to the description of how the feature helps the tapeworm survive.

[3 marks]

Feature	How the feature helps the tapeworm survive
Body sections produce many eggs	increases the rate of food absorption from the host
Hooks and suckers	protects tapeworm from damage by host's digestive enzymes
Long and flattened shape	increases the chance of infecting a new host
Thick cuticle	provides a firm grip on the intestine wall

0 2 . 3

Explain why a person infected with tapeworms might lose body mass.

[3 marks]

0 2 . 4

An obese person wants to lose weight.

Give **one** reason why the person should **not** infect themselves with a tapeworm.

[1 mark]

Turn over ►

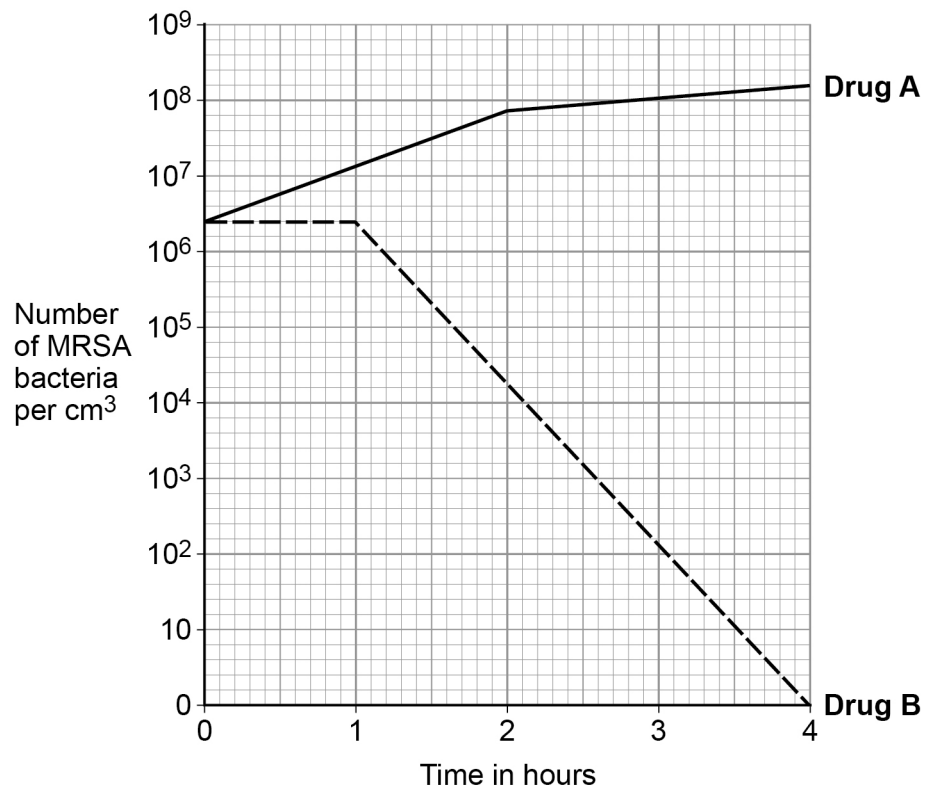


Drugs can be used to kill tapeworms. One of these drugs (**Drug B**) has been found to also kill MRSA bacteria.

An investigation compared the effects of **Drug B** with one other drug (**Drug A**) on the growth of MRSA. The MRSA bacteria were grown in nutrient broth for four hours with either **Drug A** or **Drug B**.

Figure 3 shows the results.

Figure 3



0 2 . 5

What was the dependent variable in this investigation?

[1 mark]



0 2 . 6

Describe the trends shown in **Figure 3** for **Drug A** and **Drug B**.**[4 marks]****Drug A**

Drug B

0 2 . 7

Why is it important to include a control in this type of investigation?

[1 mark]

0 2 . 8

Suggest a suitable control for this investigation.

[1 mark]

15

Turn over for the next question**Turn over ►**

0 3

Plants lose water vapour through stomata on their leaves.

Students looked at the stomata on leaves from three different plants **A**, **B** and **C**.

This is the method used.

1. Peel a thin strip from the surface of a leaf from each plant.
2. Magnify each strip and count the number of stomata.

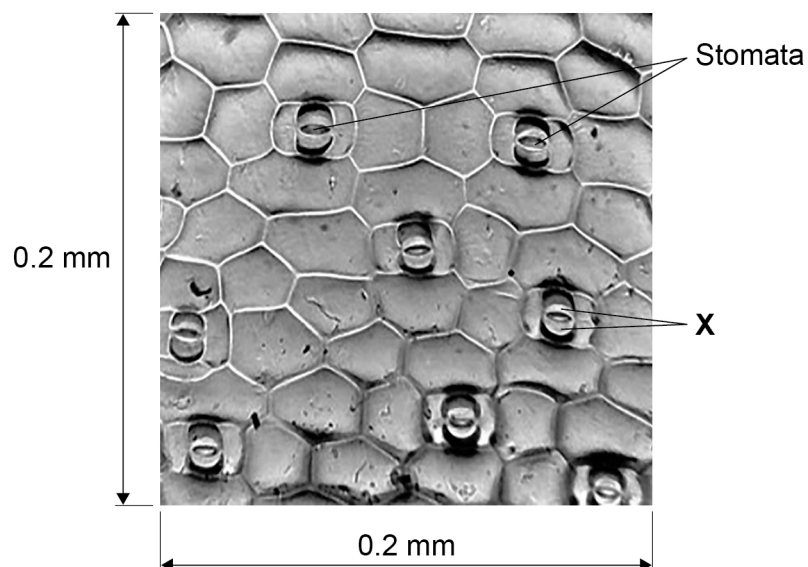
0 3 . 1

Suggest what instrument the students could use to see the stomata.

[1 mark]

Figure 4 shows stomata on the surface of a leaf.

Figure 4



0 3 . 2

Calculate the number of stomata per mm^2 on the surface of this leaf.

[2 marks]

Number of stomata = _____ per mm^2



The cells labelled **X** control the size of the stomata.

0 3 . 3 Name the cells labelled **X**.

Tick (✓) **one** box.

[1 mark]

Guard cells

☐

Palisade cells

☐

Phloem cells

☐

Xylem cells

☐

0 3 . 4 Give **two** reasons why it is important that the stomata can be opened and closed.

[2 marks]

1 _____

2 _____

Question 3 continues on the next page

Turn over ►



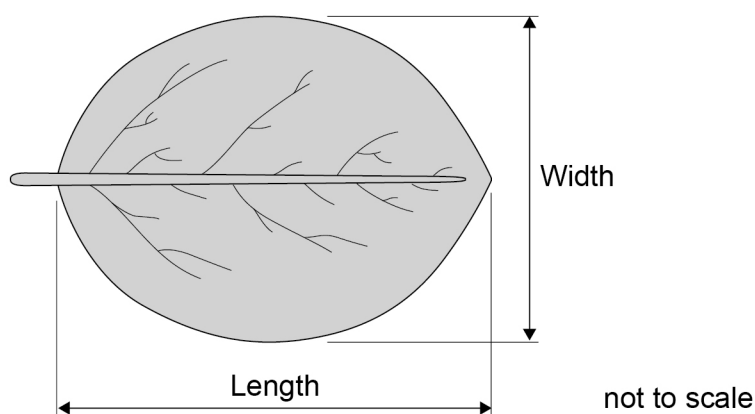
The students removed another leaf from each of the three plants **A**, **B** and **C**.

They calculated the surface area of **one** side of each leaf using two different methods.

Method 1

1. Assume the leaves are oval shaped as shown in **Figure 5**.
2. Measure the length (L) and width (W) of each leaf.

Figure 5



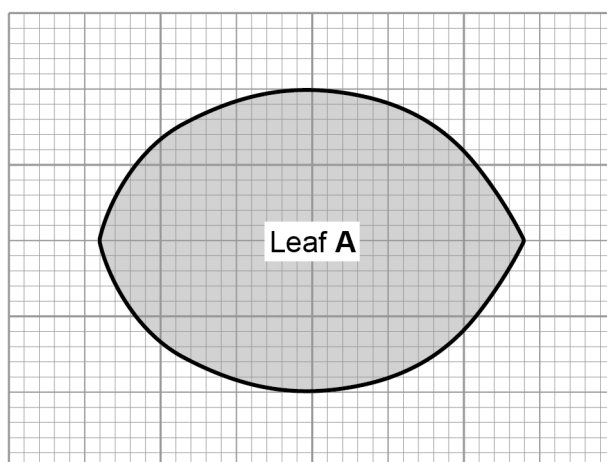
3. Calculate the surface area using the formula:

$$\frac{2 \times L \times W}{3}$$

Method 2

1. Draw around each leaf onto graph paper as shown in **Figure 6**.

Figure 6

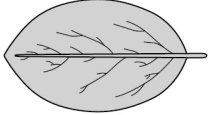
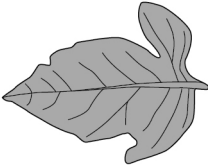
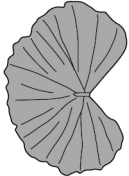


2. Count the number of squares inside each leaf shape.
3. Calculate the surface area with each square measuring 4 mm²



Table 2 shows the results. The leaves are not shown to their actual size.

Table 2

Shape of leaf	Surface area in mm ²	
	Method 1	Method 2
 A	1650	1600
 B	_____	1988
 C	1200	1760

0 3 . 5

Calculate the surface area in mm² for **one** side of leaf **B** using **Method 1**.

Use the information on page 10 and the data below to help you.

Give your answer to four significant figures.

[3 marks]

Method 1

- Length of leaf **B** = 74 mm
- Width of leaf **B** = 49 mm

Surface area = _____ mm²

Turn over ►



0 3 . 6

The students decided **not** to use the calculations for surface area obtained from **Method 1**.

Suggest **two** reasons why.

[2 marks]

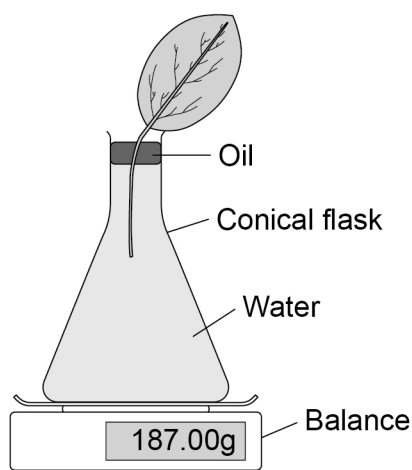
- 1 _____
- 2 _____

The students investigated whether the rate of water loss from a leaf was affected by the number of stomata. The students removed another leaf from each of the three plants, **A**, **B** and **C**.

This is the method used.

1. Calculate the total number of stomata on each leaf.
2. Set up the apparatus as shown in **Figure 7**, for each leaf.

Figure 7



3. Record the mass of each set of apparatus.
4. Leave for two days and record the new mass.
5. Calculate the difference in mass as a measure of water loss from each leaf.

0 3 . 7 What was the purpose of the layer of oil in **Figure 7**?

[1 mark]

Table 3 shows the results.

Table 3

Leaf	Surface area in mm ²	Number of stomata on top of leaf per mm ²	Number of stomata on bottom of leaf per mm ²	Total number of stomata on each leaf $\times 10^5$	Loss of water in g
A	1600	25	175	3.2	0.9
B	1988	14	122		1.4
C	1760	32	179	3.7	0.5

0 3 . 8 Calculate the total number of stomata present on leaf **B**.
Give your answer in standard form.

[3 marks]

Total number of stomata = _____ $\times 10^5$

Question 3 continues on the next page

Turn over ►



Do not write
outside the
box

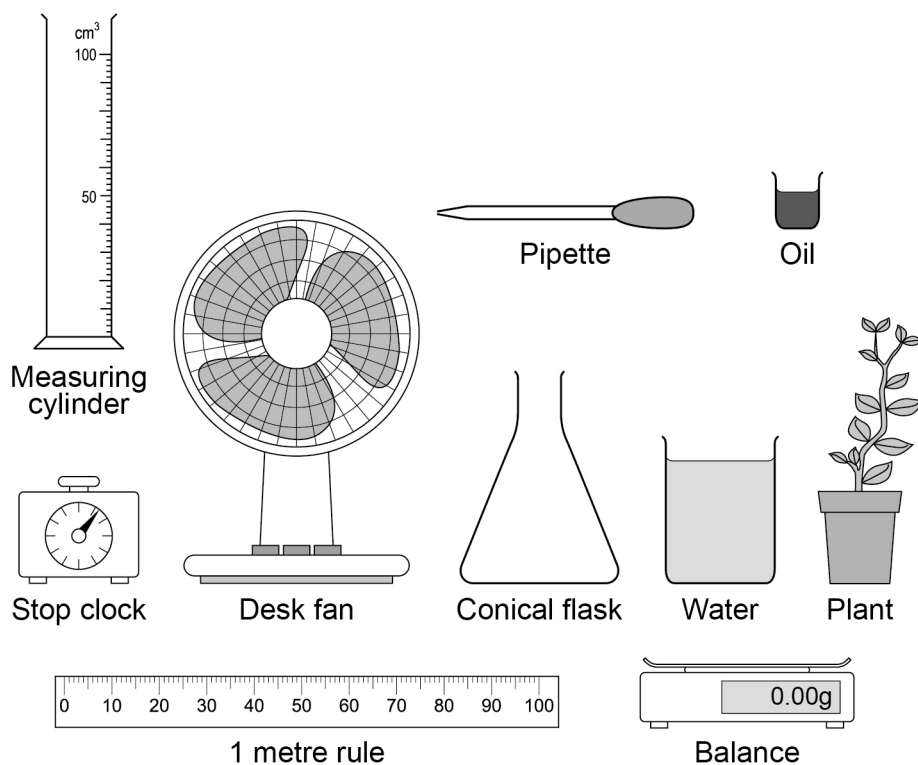
0	3	.	9
---	---	---	---

Describe how the students could investigate the effect of increasing air movement on the rate of water loss from a leaf.

[6 marks]

The students can use any of the apparatus shown in **Figure 8**.

Figure 8

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

0 4

Fats are broken down into fatty acids and glycerol during digestion.

0 4 . 1

What is the name of the enzyme that digests fats?

[1 mark]

Tick (✓) **one** box.

Amylase

☐

Hydrochloric acid

☐

Lipase

☐

Protease

☐

Some students investigated the effect of bile on the rate of fat digestion in milk. They used a pink indicator called phenolphthalein.

Phenolphthalein becomes colourless when the fat in the milk has been digested.

This is the method used.

1. Set up five test tubes as shown in **Table 4**.

Table 4

Tube number	Liquids added to tube
1	Enzyme and water
2	Bile and water
3	Enzyme and bile
4	Boiled enzyme and bile
5	Enzyme and boiled bile

- Stand all the tubes in a beaker of water at 30 °C for 10 minutes.
- Add 3 cm³ of milk and three drops of indicator to each test tube.
- Record the time taken for the indicator in the mixture to change from pink to colourless.

Turn over ►



0	4	.	2
---	---	---	---

Suggest why the tubes were left in the beaker of water for 10 minutes before adding the milk and indicator.

[1 mark]

0	4	.	3
---	---	---	---

The teacher said that the students should repeat the investigation three more times.

Give **two** reasons why it is important to repeat the investigation.

[2 marks]

1

2

0	4	.	4
---	---	---	---

The fatty acids produced during this reaction change the indicator from pink to colourless.

The students found it difficult to determine when the indicator turned colourless.

Which piece of equipment could be used instead of an indicator to measure the digestion of fat?

[1 mark]

0	4	.	5
---	---	---	---

Give **two other** ways the students could improve their investigation.

[2 marks]

1

2



The students repeated the investigation using an improved method.

Table 5 shows the students' results.

Table 5

Liquids added to test tube	Time taken to turn colourless in minutes
Enzyme and water	4
Bile and water	No change after 20 minutes
Enzyme and bile	2
Boiled enzyme and bile	
Enzyme and boiled bile	2

0 4 . 6

Predict the time taken in minutes for the mixture containing boiled enzyme and bile to turn colourless.

Explain why you made this prediction.

[3 marks]

Prediction _____

Explanation _____

Question 4 continues on the next page

Turn over ►



0 4 . 7

Give **three** conclusions about bile that can be made from the results in **Table 5**.**[3 marks]**

1 _____

2 _____

3 _____

*Do not write
outside the
box***13**

0 5

The gallbladder stores bile. Sometimes gallstones form in the gallbladder. Gallstones are made up of solid material normally formed by cholesterol and bile salts. They can be of various sizes.

0 5**1**

Table 6 shows the diameter of five gallstones.

Table 6

Diameter of gallstone in cm
0.9
1.7
1.2
2.8
2.4

Calculate the mean diameter of the gallstones.

[2 marks]

Mean = _____ cm

Question 5 continues on the next page

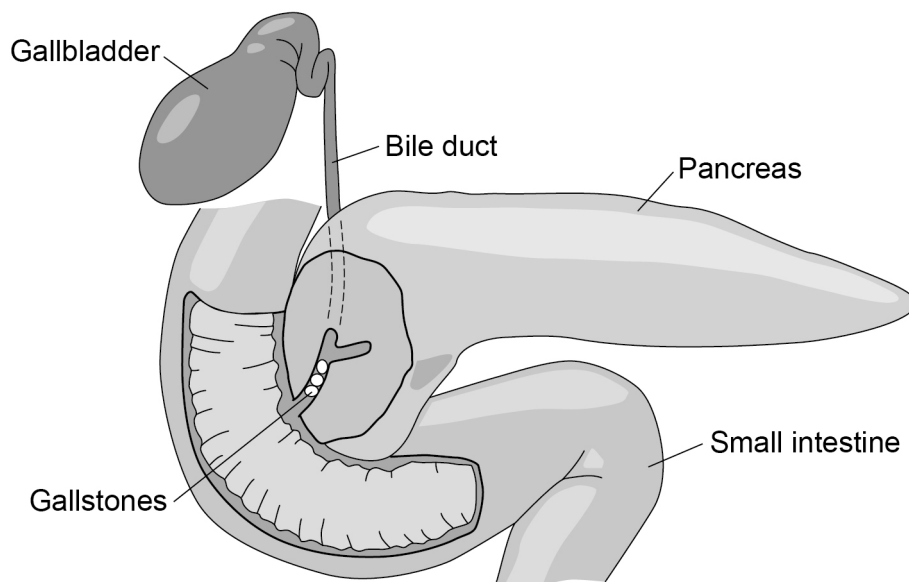
Turn over ►



Gallstones can lead to inflammation of the pancreas.

Figure 9 shows the gallbladder, pancreas and part of the small intestine of someone with a damaged pancreas.

Figure 9



Damage starts to occur when:

- gallstones move and block the bile duct
- pancreatic enzymes are forced back into the pancreas
- parts of the pancreas are destroyed by these enzymes.



Explain the effects of the damaged pancreas shown in **Figure 9** on the normal functioning of the body.

[6 marks]

[illegible]

8

Turn over for the next question

Turn over ►



0 6

Poaching and habitat loss in Africa reduced the numbers of northern white rhinoceroses from over 2000 to three in just 50 years. Two females and one male remained.

The three rhinoceroses were unable to reproduce naturally. Scientists hoped to develop techniques to breed from them.

One method is called In Vitro Fertilisation (IVF). This involves removing egg cells from the females and sperm cells from the male. A scientist injects a sperm cell into an egg cell.

0 6 . 1

Explain why the offspring produced by IVF show genetic variation.

[2 marks]

0 6 . 2

The embryos produced from IVF could be cloned. Describe the steps involved in embryo cloning.

[3 marks]

0 6 . 3

Suggest **one** advantage in cloning the embryos.

[1 mark]



0 6 . 4 A population of cloned rhinoceroses may **not** be able to survive in the wild.

Explain why.

[3 marks]

Stem cells obtained from stored tissue of 12 other northern white rhinoceroses could be used in the IVF method.

0 6 . 5 What are stem cells?

[1 mark]

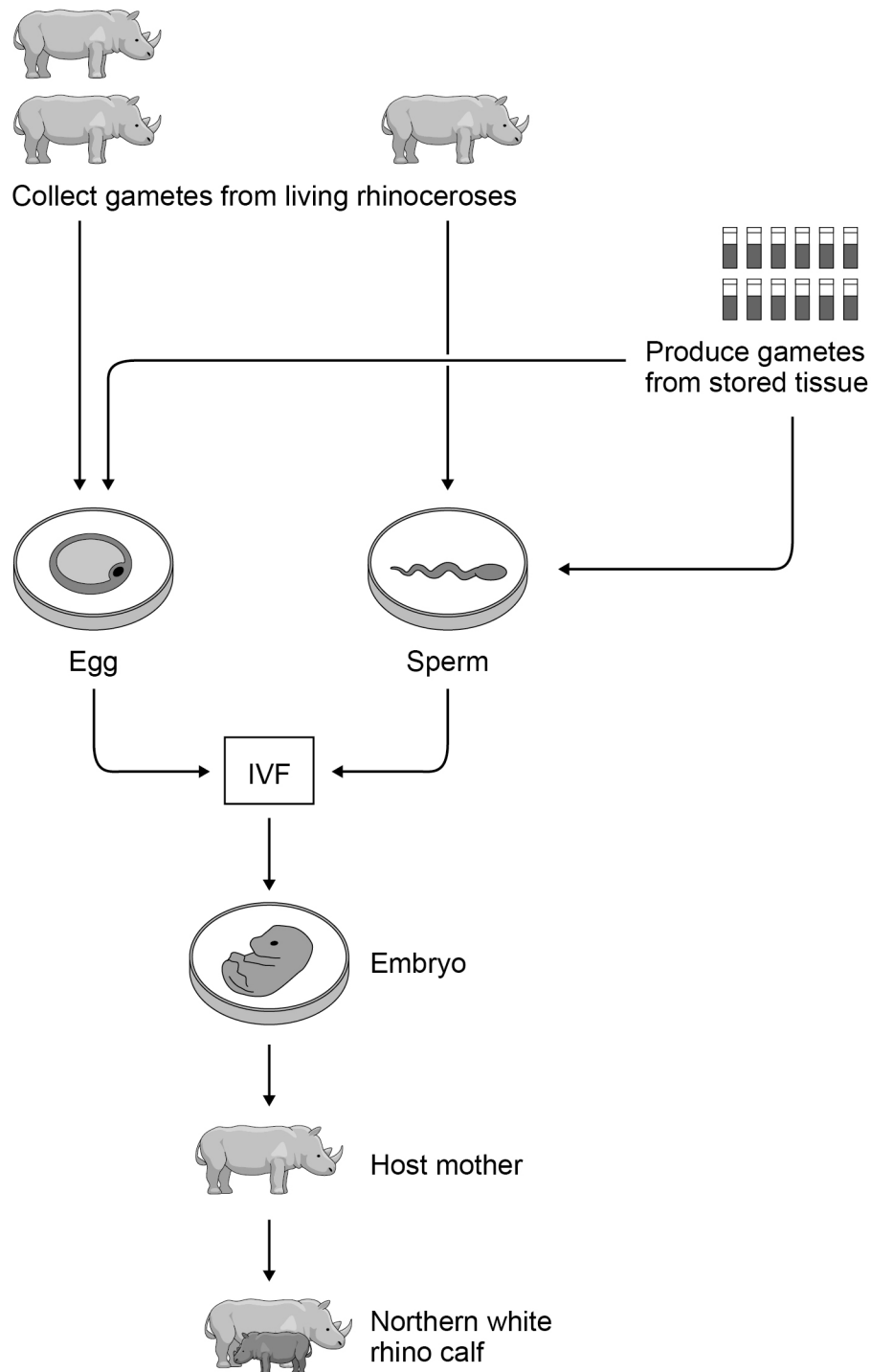
Question 6 continues on the next page

Turn over ►



Figure 10 shows a reproductive technique using IVF from natural gametes and gametes from stored tissue. This technique is expensive but has been successful with other animals. The technique will require further development before it can be used with rhinoceroses.

Figure 10



06.6

Evaluate the use of IVF to conserve northern white rhinoceroses.
You should include the use of both natural gametes and gametes from stored tissue.

[5 marks]

15**Turn over for the next question****Turn over ►**

0 7

Premature babies are babies born several weeks early.

Premature babies are at risk of hypothermia where the core body temperature falls below 35 °C. The body loses more heat than it can gain and death may occur.

0 7 . 1

There are two sets of receptors that are sensitive to temperature change.

Where are these receptors found?

[2 marks]

1 _____

2 _____

0 7 . 2

When babies are first born, their skin is wet. Wrapping premature babies in plastic film immediately after birth reduces the risk of hypothermia.

Explain how the use of plastic film reduces the risk of hypothermia.

[3 marks]



0	7	.	3
---	---	---	---

[5 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

10

END OF QUESTIONS



There are no questions printed on this page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**



[illegible]

Do not write
outside the
box

Question number	Additional page, if required. Write the question numbers in the left-hand margin.



[illegible]

There are no questions printed on this page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Copyright information

For confidentiality purposes, acknowledgements of third-party copyright material are published in a separate booklet rather than including them on the examination paper or support materials. This booklet is published after each examination series and is available for free download from www.oxfordaqaexams.org.uk after the live examination series.

Permission to reproduce all copyright material has been applied for. In some cases, efforts to contact copyright-holders may have been unsuccessful and Oxford International AQA Examinations will be happy to rectify any omissions of acknowledgements. If you have any queries please contact the Copyright Team, AQA, Stag Hill House, Guildford, GU2 7XJ.

Copyright © 2019 Oxford International AQA Examinations and its licensors. All rights reserved.

